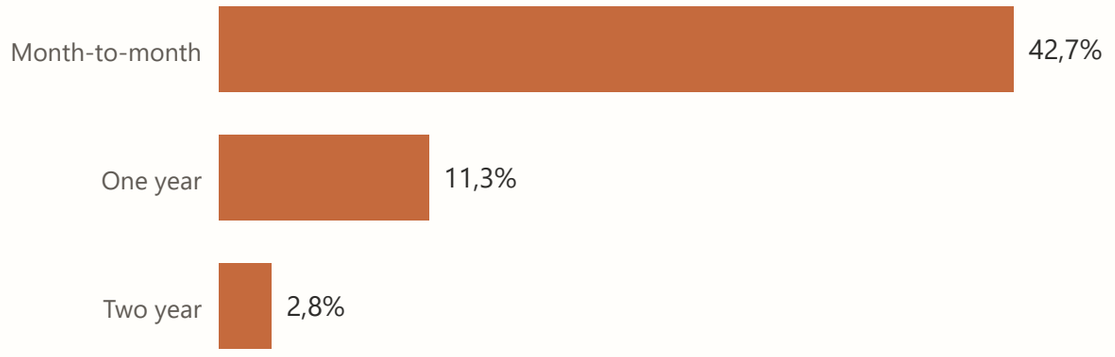
Executive churn overview

Where customer loss concentrates — before intervention

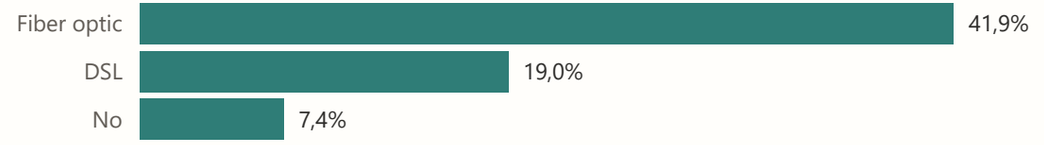
IBM TELCO / DESCRIPTIVE VIEW / ALL 7,043 AUDITED CUSTOMERS

7 043	26,54%	1 869	5 174
Total customers	Observed churn rate	Churned customers	Retained customers

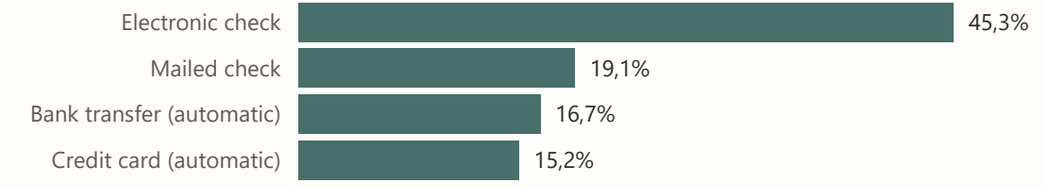
Churn by contract — the clearest divide



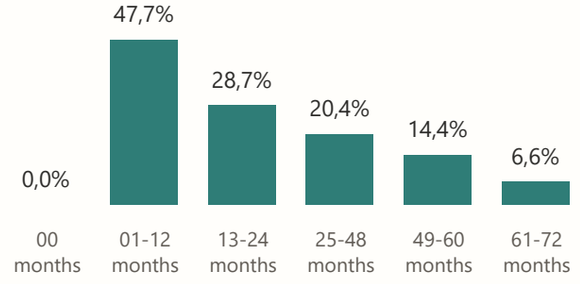
Churn by internet service



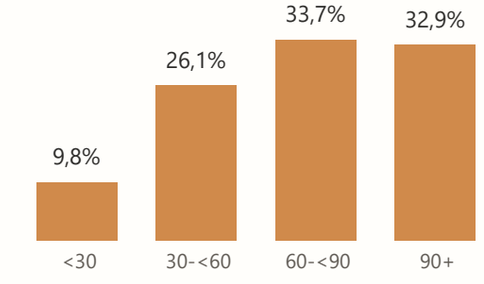
Churn by payment method



Churn by tenure band



Churn by monthly-charge band



Key findings

ALL 7,043 AUDITED CUSTOMERS

- Month-to-month churn: 42.7% vs 11.3% (one year) and 2.8% (two year).
- Months 1–12 show 47.7% observed churn; the tenure-zero group has only 11 customers.
- Fiber optic (41.9%) and electronic check (45.3%) are large, high-rate segments.

INTERPRETATION

Associations only. Segment differences do not establish causal effects, treatment uplift, or time trends.

MODEL AUDIT / FROZEN TEST

Model performance & explainability

Logistic Regression · 1,409 held-out customers

0,84385	0,65965
Held-out ROC-AUC	Held-out PR-AUC

50%	2,498
Recall @ top 20%	Lift @ top 20%

How to read this page

EVALUATION SCOPE

• 1,409 held-out customers; 374 churned (26.54%).

• Test metrics were evaluated once after training-only cross-validation.

INTERPRETATION

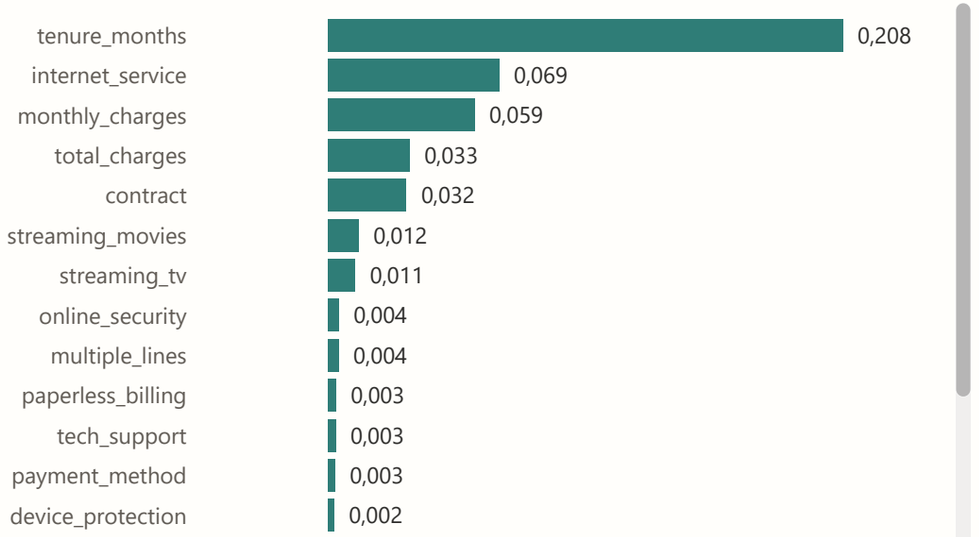
Permutation importance describes predictive associations, not causal drivers or treatment effects.

Calibration and ranking diagnostics describe one fixed random hold-out.

LIMITATION

The source has no event dates, so temporal validation and performance-drift conclusions are unavailable.

Original-feature permutation importance



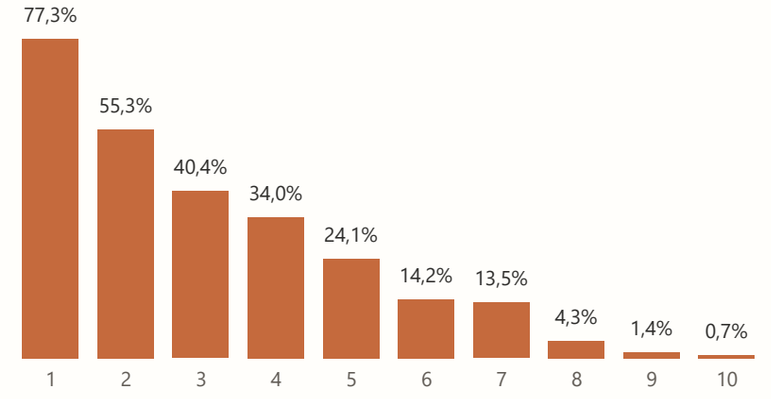
Training-only 5-fold cross-validation



Confusion matrix · illustrative threshold 0.50

Confusion TN	Confusion FP	Confusion FN	Confusion TP
920	115	175	199

Observed churn by risk decile · 1 = highest risk



Calibration by equal-count decile · ECE 2.44%

Risk decile	Customers	Predicted risk	Observed churn	Absolute gap
1	141	72,0%	77,3%	5,3%
2	141	59,5%	55,3%	4,1%
3	141	46,2%	40,4%	5,8%
4	141	33,5%	34,0%	0,5%
5	141	22,9%	24,1%	1,2%
6	141	15,4%	14,2%	1,2%
7	141	8,1%	13,5%	5,4%
8	141	3,9%	4,3%	0,3%
9	141	1,6%	1,4%	0,2%
10	140	0,5%	0,7%	0,3%

RETENTION PLANNING / HYPOTHETICAL SCENARIO

Retention strategy & scenario simulator

Probability-weighted decision support · frozen 1,409-customer held-out risk ranking

SCENARIO INPUTS

Six disconnected what-if controls

Base-case reference: 20% capacity · 20% effect · 55% margin · 6 mo · costs 5 + 5

Contact capacity	Effectiveness
20%	20%
Contribution margin	**Benefit horizon · mo…**
55%	6
Contact cost · units	**Offer cost · units**
5	5

DECISION BOUNDARIES

How to read this page

• Dynamic outputs use frozen predicted probabilities and selected hypothetical inputs—not actual churn outcomes.
• Effectiveness is a relative reduction in predicted churn risk. Contact and offer costs are incurred once per contacted customer.
• Monetary outputs are illustrative currency units, not actual profit, verified ROI, or customer lifetime value.
• Predictions do not estimate causal treatment uplift. Retrospective capacity metrics use held-out labels only in aggregate; no temporal validation was performed.

7 022
Net expected value · illustrative currency units

9 842
Gross retained contribution · units

2 820
Campaign cost · units

282
Contacted customers

37,1
Expected prevented churn · customers

5,7%
Break-even effectiveness · relative risk reduction

Retrospective targeting performance

Frozen held-out test · aggregate label-based evaluation

Capacity	Contacted	Churn captured	Precision	Lift
Top 10%	141	29,1%	77,3%	2,912
Top 20%	282	50,0%	66,3%	2,498
Top 30%	423	65,2%	57,7%	2,173

RETROSPECTIVE HELD-OUT EVALUATION

Top 20% targeting captures 50% of observed churn, with 2.50x lift.

Frozen 1,409-customer test set. This is aggregate retrospective evaluation—not a measured retention-campaign outcome.